RAHUL SAHANI

AI ENGINEER | PYTHON | LLM & ML APPLICATIONS

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PROFESSIONAL SUMMARY

AI-focused Python developer with project experience building LLM-powered applications, retrieval-augmented generation (RAG), machine learning APIs, NLP workflows, and AI-enabled automation. Built a document-grounded chatbot with persistent vector retrieval and source attribution, integrated Gemini text and vision models, and developed a FastAPI recommendation service using PyTorch Geometric. Interested in building reliable AI services, data pipelines, evaluation workflows, and developer tools.

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript, Git, GitHub
Machine Learning & NLP: PyTorch, PyTorch Geometric, TensorFlow/Keras, scikit-learn, NLTK/VADER, TF-IDF, cosine similarity, basic model evaluation
LLMs & APIs: Google Gemini text/vision API, FastAPI, Streamlit, REST/JSON APIs, environment-based configuration
Data Pipelines: Pandas, NumPy, Selenium, BeautifulSoup, text cleaning/tokenization, dataset processing, checkpoint and retry workflows
Tools & Cloud: Jupyter Notebook, AWS EC2, GCP, debugging, API integration

PROFESSIONAL EXPERIENCE

Software Developer | Printeast | Apr 2026 - Jun 2026
- Contributed to frontend and full-stack web development, implementing, integrating, and debugging application features.

SELECTED AI & AUTOMATION PROJECTS

YouTube AI Upload Automation | Python, Gemini, NVIDIA NIM, DeepSeek, Streamlit, SQLite, FFmpeg, YouTube Data API
- Developed an end-to-end video publishing pipeline for audio extraction, Gemini transcription, AI-generated titles/descriptions/tags, OAuth-authenticated resumable YouTube uploads, and automated publishing. Built a Streamlit dashboard for pipeline progress, execution history, success/failure metrics, and published-video analytics; generated metadata concurrently with video uploads.

RAG Document Chatbot | Python, Gemini 2.5 Flash, ChromaDB, Sentence Transformers, Streamlit
- Built a document question-answering pipeline for PDF/TXT files with extraction, overlapping text chunks, local embeddings, persistent vector search, hash-based change detection, Gemini response generation, and source attribution.

Gemini Pro Vision Chatbot | Python, Streamlit, Google Gemini, Pillow
- Built a conversational image-analysis app integrating Gemini text and vision models, image uploads, image encoding, and session-based chat history.

Graph-Based Book Recommendation API | Python, FastAPI, PyTorch Geometric, Pandas, scikit-learn
- Implemented a graph-convolution model and FastAPI endpoints combining graph-based recommendations with TF-IDF/cosine-similarity and popularity-based recommendations.

Movie Review Sentiment Analysis | Python, NLTK/VADER, Pandas, Selenium, BeautifulSoup
- Built an IMDb review collection and NLP preprocessing workflow with text cleaning, tokenization, stopword removal, and lemmatization; applied VADER sentiment scoring and generated distribution and timeline visualizations.

Parking Spot Occupancy Detection | Python, TensorFlow/Keras, OpenCV, NumPy
- Combined CNN transfer learning with OpenCV-based parking-space detection to classify occupancy from parking imagery and video.

EDUCATION

Master of Computer Applications (MCA) | 2024 - 2026
Bachelor of Computer Applications (BCA) | Institute of Information Technology and Management | 2021 - 2024

CERTIFICATIONS & ACHIEVEMENTS

Wipro Java J2EE Certification; 1st Place, ODE TO CODE (Mar 2024); 1st Place, CSS Battle, MATRIX Internware IITM (Oct 2023); International Space Search Collaboration asteroid discovery (Aug 2022).